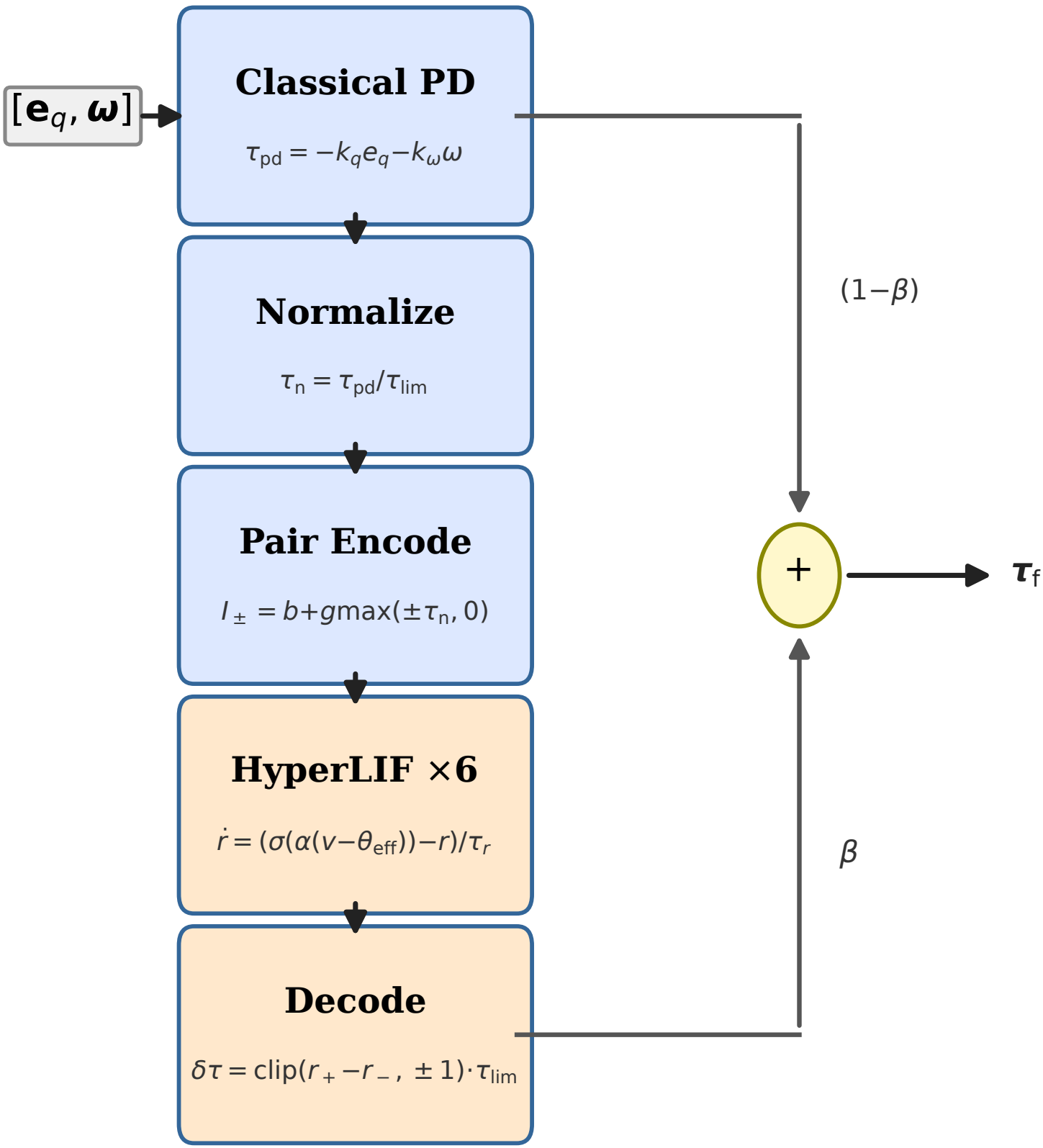
Architecture A — Body-Torque Residual
$[\mathbf{e}_q, \boldsymbol{\omega}]$
Classical PD
$\tau_{\text{pd}} = -k_q e_q - k_\omega \omega$
Normalize
$\tau_{\text{n}} = \tau_{\text{pd}} / \tau_{\text{lim}}$
Pair Encode
$I_{\pm} = b + g\max(\pm\tau_{\text{n}}, 0)$
HyperLIF ×6
$\dot{r} = (\sigma(\alpha(v - \theta_{\text{eff}})) - r)/\tau_r$
Decode
$\delta\tau = \text{clip}(r_+ - r_-, \pm 1) \cdot \tau_{\text{lim}}$
$(1-\beta)$
β
+
$\boldsymbol{\tau}_{\text{f}}$